

Understand scalability in Power BI



Learning objectives

- Describe the importance of building scalable semantic models
- Implement Power BI data modeling best practices
- Use the Power BI large dataset storage format

Enterprise-scale data: defined

Massive

Complex

Diverse

Rapid

Valuable

What is scalability and why is it important?

Flexibility

Models need to be able to accommodate change

Data growth

Models must be able to handle an increase in data volume with acceptable report performance

Reduced complexity

Models built with scalability in mind will be less complex and easier to manage

How do I design for scalability?

- Build with data modeling best practices in mind
- Determine your needs for Direct Lake/ Import/ Direct Query
- Enable large semantic model storage feature

Implement data modeling best practices

Choose the correct model framework:

- Import
- DirectQuery
- Direct Lake

Implement data modeling best practices:

- Do as much data preparation work as possible before data reaches Power BI, as far upstream as possible.

Use the large semantic model feature

Enable large dataset storage format:

- Semantic model sizes limited only by the capacity size or a maximum size set by the admin
- Can be enabled for all Premium SKUs

Large semantic model storage format

For most Premium capacities, using large semantic model storage format can improve performance. [Learn more](#)



On

Apply

Discard

Power BI datasets store data in a highly compressed, in-memory cache for optimized query performance. Enterprise deployment of an analytics solution using Power BI will likely require Power BI Premium. With the *large dataset storage format* enabled, dataset sizes are limited only by the capacity size, or a maximum size set by the administrator. This differs from datasets in Power BI Premium, which are limited to 10 GB after compression if large dataset storage format isn't enabled.

Large semantic models can be enabled for all Premium P SKUs, Embedded A SKUs, and with Premium Per User (PPU). The large dataset size limit in Premium is comparable to Azure Analysis Services, in terms of data model size limitations. Enabling the large semantic model format enables fast user interactivity and allows data to grow beyond the 10-GB limit.

To use large semantic model storage format, **the dataset must be stored in a workspace that allocated to Premium capacity.**

Important note: Datasets enabled for large models can't be downloaded as a Power BI Desktop (.pbix) file from the Power BI service.

Configure Large Semantic Models

Can also enable large dataset storage format for all semantic models created in a workspace

- You can set the default storage format for all semantic models created in a workspace in the workspace settings.

Default storage format

☐ Small semantic model storage format

☒ Large semantic model storage format

[Learn more about semantic model storage formats](#) 

Apply

Cancel

Knowledge check



- 1 What is the most efficient approach for data transformation when designing semantic models for scalability?**
 - ☐ Data transformation should always be done using Power Query in Power BI.
 - ☒ Data transformation should be done as close to the source as possible, before reaching Power BI.
 - ☐ Data transformation should be done using DAX, after data is ingested and loaded to the model.
- 2 Which of the following best practices for Power BI data modeling is relevant only to DirectQuery models?**
 - ☒ Set relationships to enforce integrity using the assume referential integrity property on relationships.
 - ☐ Use a star schema as opposed to wide tables.
 - ☐ Disable auto date/time in Power BI Desktop.
- 3 What are the size limits on a semantic model with large semantic model storage format determined by?**
 - ☐ Power BI 10 GB size limit.
 - ☒ Fabric capacity size or the maximum size set by the administrator.
 - ☐ There are no size limits if large dataset storage format is enabled.

Recap

In this section, we learned:

- Scalability and working with large data is achievable in Power BI.
- The correct model framework, a proper data model, and using Fabric features like large dataset storage enables performant enterprise reporting in Power BI.

Use tools to optimize Power BI performance



Learning objectives



- Optimize solutions using aggregations, dual storage mode, and hybrid tables.
- Optimize queries using performance analyzer.
- Troubleshoot DAX performance using DAX Studio.
- Optimize a data model using Tabular Editor.

Optimize Power BI solutions



Data source(s)



Data model



Visualizations



Environment

You can optimize your solution at different architectural layers. Layers include:

- The data source(s)
- The data model
- Visualizations, including dashboards, Power BI reports, and Power BI paginated reports
- The environment, including capacities, data gateways, and the network

The data model supports the entire visualization experience. Data models are either external-hosted or internal-hosted, and in Power BI they are referred to as *datasets*. It's important to understand your options, and to choose the appropriate dataset type for your solution: import v DirectQuery v Composite.

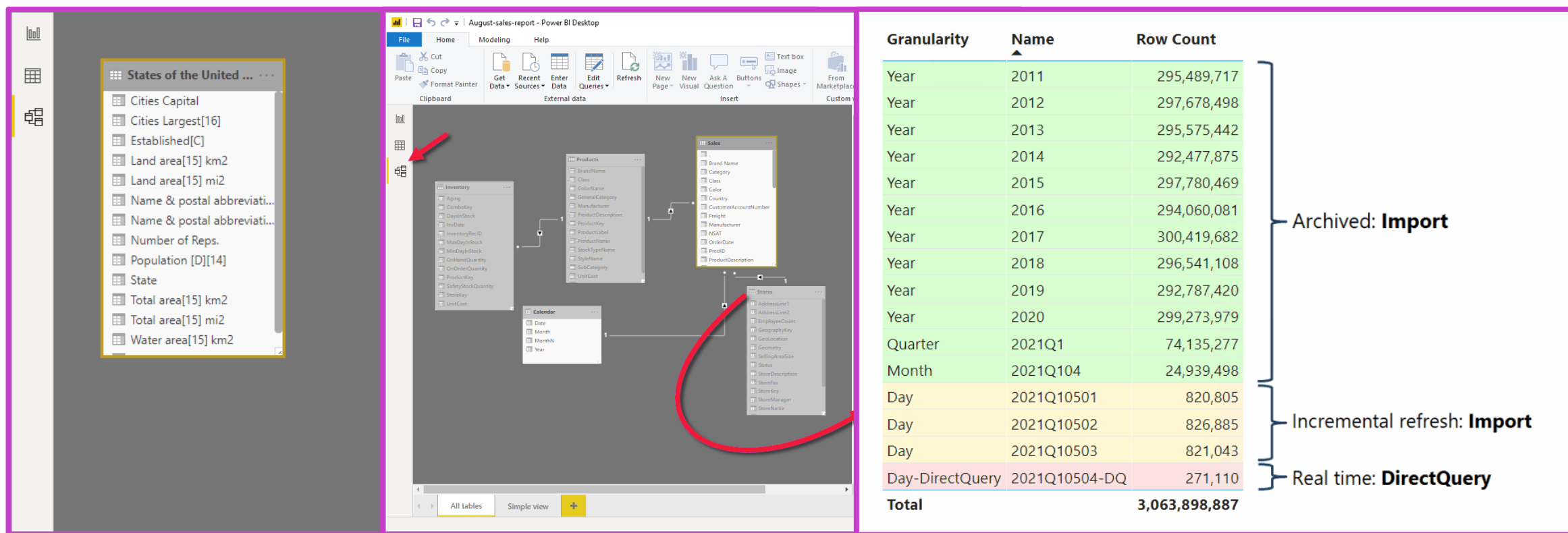
You can optimize the Power BI environment by configuring capacity settings, sizing data gateways, and reducing network latency.

Optimize Power BI solutions

Aggregations

Dual storage mode

Hybrid tables



Aggregations

- Aggregations in Power BI can improve query performance over very large DirectQuery datasets. By using aggregations, you cache data at the aggregated level in-memory. Aggregations in Power BI can be manually configured in the data model, as described in this article, or for Premium subscriptions, automatically by enabling the Automatic aggregations feature in dataset Settings.

Dual storage mode

In Power BI Desktop, you can specify the storage mode of a table. The storage mode lets you control whether Power BI Desktop caches table data in-memory for reports. Setting the storage mode provides many advantages. You can set the storage mode for each table individually in your model.

Benefits:

- **Query performance:** As users interact with visuals in Power BI reports, Data Analysis Expressions (DAX) queries are submitted to the dataset. Caching data into memory by properly setting the storage mode can boost the query performance and interactivity of your reports.
- **Large datasets:** Tables that aren't cached don't consume memory for caching purposes. You can enable interactive analysis over large datasets that are too large or expensive to completely cache into memory. You can choose which tables are worth caching, and which aren't.
- **Data refresh optimization:** You don't need to refresh tables that aren't cached. You can reduce refresh times by caching only the data that's necessary to meet your service level agreements and your business requirements.
- **Near-real time requirements:** Tables with near-real time requirements might benefit from not being cached, to reduce data latency.
- **Writeback:** Writeback enables business users to explore what-if scenarios by changing cell values. Custom applications can apply changes to the data source. Tables that aren't cached can display changes immediately, which allows instant analysis of the effects.

Set the **Storage mode** property to one of these three values:

- **Import:** Imported tables with this setting are cached. Queries submitted to the Power BI dataset that return data from Import tables can be fulfilled only from cached data.
- **DirectQuery:** Tables with this setting aren't cached. Queries that you submit to the Power BI dataset—for example, DAX queries—and that return data from DirectQuery tables can be fulfilled only by executing on-demand queries to the data source. Queries that you submit to the data source use the query language for that data source, for example, SQL.
- **Dual:** Tables with this setting can act as either cached or not cached, depending on the context of the query that's submitted to the Power BI dataset. In some cases, you fulfill queries from cached data. In other cases, you fulfill queries by executing an on-demand query to the data source.
- Changing the **Storage mode** of a table to **Import** is an *irreversible* operation. Once set, this property can't later be changed to either **DirectQuery** or **Dual**.

Hybrid tables

- A hybrid table can help to strike the right balance between query performance and data freshness.
- A hybrid table is essentially a (large) table that has one or multiple import-mode partitions as well as another partition in DirectQuery mode.
- If the DirectQuery partition is sufficiently small in comparison to the import-mode partitions, the query/response roundtrips between Power BI and the data source should still be reasonably fast while access to the bulk of the data is already super-fast in import mode. Import and DirectQuery data is presented to users as a single table with business definitions and calculations.

Optimize Power BI models using tools

Retroactive

**Performance
Analyzer**

DAX Studio

Proactive

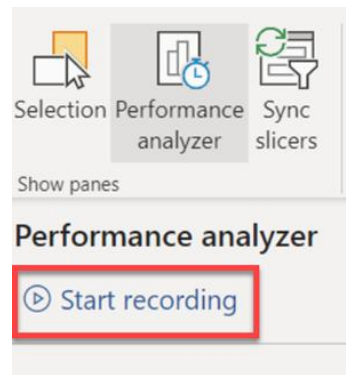
**Best Practice
Analyzer in
Tabular Editor**

You've built Power BI reports and they're running slow. How do you identify the source of the problem? The report could be slow because of issues at the *source*, the structure of the *data model*, the *visuals* on the report page, or the *environment*.

Optimizing performance in the context of this section means making changes to the *data model* so that it runs more efficiently. The tools we're discussing in this section will help you troubleshoot and improve the data model, which ultimately improves user experience.

Ideally, you should assess performance as you move through the stages of the development lifecycle; development, testing, production, and optimization of your solutions.

Use Performance analyzer



Performance analyzer

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms)
Recording started (7/13/2022 3:18:53 PM)	-
1 Changed page	-
Slicer	378
Slicer	376
Slicer	377
Table	298
2 Changed a slicer	-
Slicer	45
Slicer	246
Slicer	388
Table	520
DAX query	34
Evaluated parameters	28
Visual display	131
Other	327
Copy query	
3 Cross-highlighted	-
Slicer	149
Slicer	148
Slicer	147

Performance Analyzer

The Performance analyzer helps you optimize at two of the four architecture levels, the *data model* and *report visuals*. The Performance analyzer is a great place to start when you're optimizing reports.

The Performance analyzer displays and records logs that measure how each of your report elements performs when users interact with them. You can also see which aspects of their performance are most (or least) resource intensive. You can immediately see results, make changes, run the analyzer again, and see the impact of changes you've made.

After you start recording, take actions in the report. For example, move from one report tab to the next, select a slicer item, or interact with any of your visuals. Any actions you take in the report are displayed and logged in real time in the Performance Analyzer pane, in the order that the visual is loaded by Power BI.

The Performance analyzer is looking at the time it takes for each visual to query the data model and render results. This is the time from when a user does something on the page to when the visual is rendered, in milliseconds.

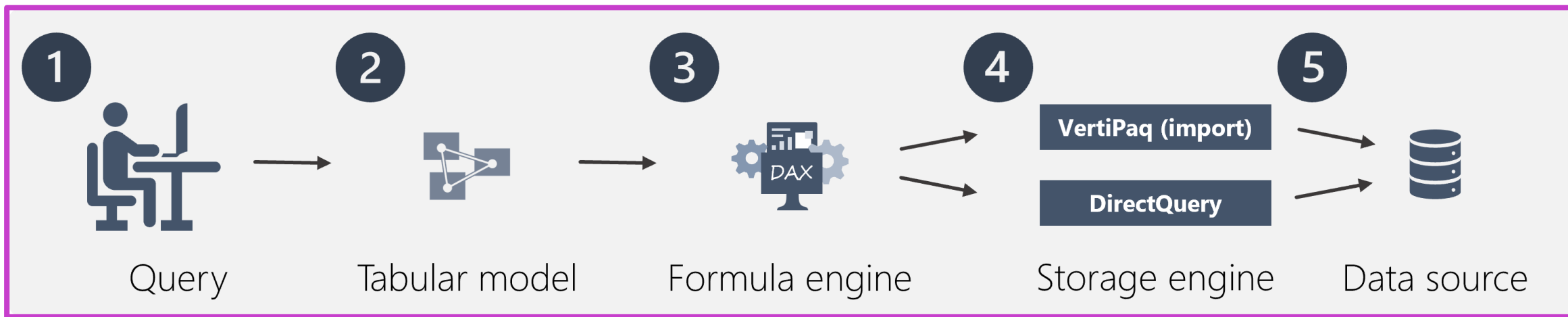
In the example on the slide, three actions were taken after recording started.

1. Navigated to a new report page
2. Changed a slicer value
3. Cross highlighted a table

As users interact with visuals in Power BI reports, DAX queries are submitted to the dataset and cached into memory. Because of this, you may need to clear the DAX query cache to get accurate results in the Performance analyzer. You can clear the cache by either closing and re-opening your report, or using DAX Studio.

Use DAX Studio

Understand the VertiPaq engine



DAX Studio

DAX Studio is an open-source tool for executing DAX queries against Power BI and Analysis Services models. DAX Studio is useful for:

- Reviewing the contents of your data model.
- Writing and optimizing complex DAX formulas and queries.

Before jumping into optimization, it's important to understand what's happening under the hood of Power BI. By using compression algorithms and a multi-threaded query processor, the Analysis Services *VertiPaq engine* delivers fast access to tabular model objects and data by Power BI.

Power BI reads the content of your data source and transforms it in the internal VertiPaq columnar data structure, where each column is encoded and compressed. Dictionaries and indexes are created for each column. Finally, data structures are created for relationships and computation and compression of calculated columns occurs.

DAX queries are being processed by two engines, the *formula engine* and the *storage engine*.

Calculations using DAX, either measures or columns, are a part of a DAX query, which is processed by two engines in VertiPaq. When a query is processed, the *formula engine* processes the request, asks the storage engine for data, and performs necessary calculations.

The *storage engine* retrieves and aggregates data requested by the formula engine.

In the diagram on screen, the DAX query is sent to the tabular model in steps 1 and 2. The request is then processed by the formula engine and sent to the storage engine, represented by step 3. In step 4, the storage engine either retrieves the data from the model and stores it in memory (for import mode), or passes the query on to the data source (for DirectQuery). For import mode, refreshing the data will retrieve the data from the source.

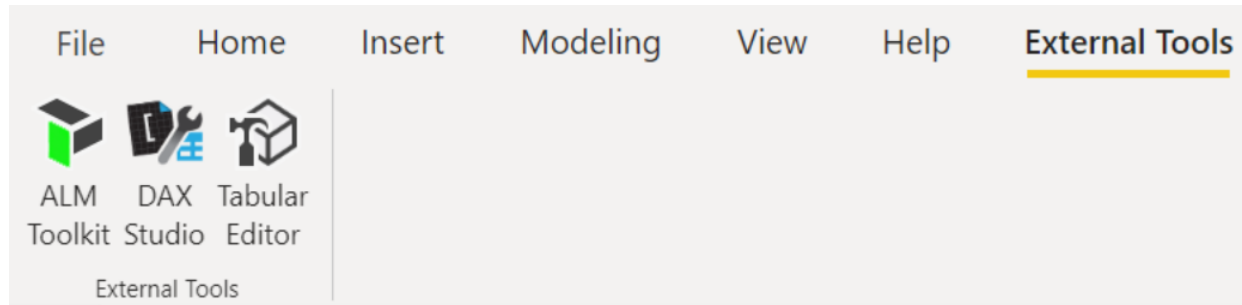
Use DAX Studio

View model metrics using the VertiPaq Analyzer

VertiPaq Analyzer Metrics

Tables	Columns	Relationships	Partitions	Summary										
Name	Cardinality	Table Size	Col Size	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table	% DB	
trips	27,514,229	3,968,498,7...	3,889,411,4...	773,132,304	2,608,051,5...	508,227,592	Many	-	-	0	79,087,328		99.58%	
End date	25,013,372	3,968,498,784	1,570,303,664	75,104,528	1,295,092,112	200,107,024	HASH	DateTime	-	-		40.37%	39.40%	
Start date	24,206,001	3,968,498,784	1,551,559,176	71,432,192	1,286,478,936	193,648,048	HASH	DateTime	-	-		39.89%	38.93%	
TripID	27,514,229	3,968,498,784	183,428,496	73,371,424	136	110,056,936	VALUE	Double	-	-	-	4.72%	4.60%	

Optimize data model by using Best Practice Analyzer



The best practice analyzer scans your model for issues whenever a change is made to the model.

The image shows the 'Best Practice Analyzer' window. It displays a list of violations categorized by rule. The table below summarizes the violations shown in the screenshot.

Object	Type
Do not use floating point data types (2 objects)	
'Product'[Weight]	Column
'Sales'[Exchange Rate]	Column
Don't summarize numeric columns (4 objects)	
'Product'[Unit Cost]	Column
'Product'[Unit Price]	Column
'Product'[Product Sales]	Column
'Product'[Product Sales Amount]	Column
Organize columns and hierarchies in display folders (2 objects)	
'Customer'	Table (DQ ov
'Product'	Table (DQ ov
Hide foreign key columns (1 object)	
'Product'[Brand]	Column
Avoid CamelCase on visible columns and hierarchies (1 object)	
'Store'[StoreKey]	Column
Avoid CamelCase on visible measures and tables	
No property or field 'Table' exists in type 'Table'	Error
Remove unused columns (9 objects)	
'Date'[Working Day Number]	Column
'Sales'[Order Number]	Column
'Sales'[Line Number]	Column
'Sales'[Quantity]	Column
'Sales'[Unit Price]	Column
'Sales'[Net Price]	Column
'Sales'[Unit Cost]	Column
'Sales'[Currency Code]	Column
'Sales'[Exchange Rate]	Column

20 objects in violation of 7 Best Practice rules.

Tabular Editor is an alternative tool for authoring tabular models for Analysis Services and Power BI. Tabular Editor 2 is an open-source project that can edit a BIM file without accessing any data in the model. Tabular editor enables users to easily manipulate tabular models and can be used in a few different ways. Tabular Editor can be used to:

- quickly edit data models
- automate repetitive tasks using scripting
- incorporate DevOps with tabular models
- run BPA rules

BPA is a set of rules run in Tabular Editor that notify you of potential modeling missteps or changes that you can make to improve your model design and performance. It includes recommendations for naming, user experience, and common optimizations that you can apply to improve performance.

To use the pre-defined BPA rules, download the [BPA rules.json](#) file from the GitHub repository. The rules are divided into categories for easier viewing, for example:

- Performance
- DAX Expressions
- Error Prevention
- Formatting
- Maintenance

BPA was designed to enable you to create custom rules and best practices for your organization. If the Best Practice Rules don't suit your needs, you can create rules required for data modeling best practices as you deem appropriate.

Exercise



aka.ms/mslearn-fabric-optimize-pbi-performance

Use tools to optimize Power BI performance

- The estimated time to complete this exercise is **45 minutes**.

Knowledge check (1/2)



1 Stephanie is an enterprise data analyst, working as a resource to data analysts sitting in the finance department. One of the finance analysts asked Stephanie to help troubleshoot why report visuals take 5 seconds to render after a slicer selection is made. What tool should Stephanie use first to begin troubleshooting?

- ☐ Visual Studio.
- ☐ DAX Studio.
- ☒ Performance analyzer.

2 James has been asked to troubleshoot a report that contains a matrix visual that renders visibly slower than any other visuals on the page. After running the Performance analyzer, James notices that he needs to dig further into the DAX query for the matrix in DAX Studio to understand what's causing the trouble. That same query is running in less than a second in DAX studio. Has James solved the problem?

- ☐ Yes. James doesn't need to make any more changes to the DAX query.
- ☐ No. James needs to use Tabular Editor to continue to investigate.
- ☒ No. James needs to clear the model cache to ensure that query results aren't cached in the model.

Knowledge check (2/2)



3 Mary-Jo is responsible for managing the maintenance and deployment of Power BI assets for her entire organization. What tool can Mary-Jo use to ensure all data models adhere to data modeling best practices?

- ☐ DAX Studio.
- ☒ Best Practice Analyzer in Tabular Editor.
- ☐ Performance analyzer.

Recap

In this section, we:

- Created Power BI model relationships
- Reviewed DAX concepts
- Created calculation groups
- Enforced Power BI model security
- Used tools to optimize Power BI performance

Further reading

Use tools to optimize Power BI performance

<https://aka.ms/optimize-power-bi>

Optimization guide for Power BI

<https://learn.microsoft.com/power-bi/guidance/power-bi-optimization>



Feedback Form

